ADL 2026 HW2 report: product retrieval for a sales chatbot

Code and logs: https://github.com/goog-msft-fb-nflx-nvda-aapl/csie5431-adl-hw2 (private until after the deadline, made public afterwards; `docs/WORKLOG.md` there is the full experiment log).

All numbers are on the public split (1,000 dialogues, 75,000 products) unless stated. "Graded" = the TA's `evaluate.py` (gold or judge-accepted alternative counts, judge Qwen2.5-1.5B-Instruct); "exact" = gold ID only. Timings are from a shared NVIDIA H200; **no RTX 3070 was available, so 3070 speed was not measured.** Hyperparameters were chosen using public metrics (see Limitations).

Q1. Retrieval and reranking strategy

Submitted pipeline

1. **Dense retrieval.** Qwen3-Embedding-0.6B fine-tuned on the training dialogues. The query is the TA's all-turn text (role-tagged turns, as built by `build_eval_query`) with the model's own query instruction prompt; products are `title\ndescription`, truncated to 384 tokens.
2. **BM25** (scikit-learn CountVectorizer, English stop words, k1 1.2, b 0.75) over the same text.
3. **Fusion**: reciprocal rank fusion (k = 60) of the two top-200 lists.
4. **Rerank** the top 30 with Qwen3-Reranker-0.6B fine-tuned (score = logit("yes") - logit("no") at the last position, fp16, 1,400 characters of product text per candidate), return the top 10.
5. **Query-likelihood rescoring** (from the lecture's query-likelihood slides; zero-shot, no training): Qwen2.5-1.5B-Instruct computes the mean log-probability of the conversation text given the product listing for the reranker's top 15; this is z-scored, added to the z-scored reranker score within those 15 and the 15 are re-sorted. It is an optional last stage: skipped for queries when time is short, and skipped altogether if the model cannot be loaded.
6. A wall-clock guard shrinks the dense max length and the rerank depth if the measured speed projects past a 3,000 s budget (reranking has priority, query likelihood only uses the time left), so a slower GPU degrades quality instead of exceeding the one-hour limit.

Training data construction

No external dataset is used; the only extra supervision is produced by the provided judge model.

Dense retriever. Anchor = conversation text; positive = the gold product text; negatives = the other positives in the batch (batch 128) plus **3 mined hard negatives per query**, sampled uniformly from ranks 5-50 of the zero-shot BM25 + Qwen3-Embedding hybrid ranking, excluding the gold and any product with the same title as the gold (colour/size variants). Skipping ranks 1-4 is a precaution against false negatives (acceptable alternatives would count as correct at test time). 4,830 training dialogues (all of them for the submitted model).

Reranker. For each training dialogue the top-20 zero-shot hybrid candidates were judged with the official judge (gold as reference), 93,198 judgments: 8.2% of non-gold candidates were accepted and 54.8% of dialogues have at least one accepted alternative in their top 20. A training group is 1 positive + 7 negatives: the positive is the gold, or with probability 0.5 a random judge-accepted alternative (when one exists); negatives are sampled from the judge-rejected candidates. This keeps probable false negatives out of the negatives and teaches the reranker the same notion of relevance the metric uses. The labels (`external/judge_labels_train.json`, 1.7MB, keyed by dialogue id and product id) took about 2 h 20 min to produce on a shared H200, so `download.sh` ships the file that `judge_labels.py` wrote and `run.sh` regenerates it only if it is absent.

Loss and hyperparameters

- Dense: `CachedMultipleNegativesRankingLoss` (InfoNCE over cosine similarity, scale 20, i.e. temperature 0.05, in-batch + hard negatives, GradCache mini-batch of 8 texts), batch 128, lr 3e-5, 2 epochs (76 steps), linear decay with 10% warm-up, bf16, gradient checkpointing, training max length 384, `NO_DUPLICATES` batch sampler.
- Reranker: listwise softmax cross-entropy over the 8 scores of a group with the positive as target, full fine-tune, AdamW lr 2e-5, one dialogue per micro-step with gradients accumulated over 2 dialogues, 1 epoch (2,415 optimizer steps), linear decay with 10% warm-up, bf16, gradient checkpointing, max length 1,024, 1,400 characters per product.
- Base models: `Qwen/Qwen3-Embedding-0.6B`, `Qwen/Qwen3-Reranker-0.6B`.

Training fits the 8GB GPU

The TA stated that `run.sh` will be run and that training must fit the RTX 3070's 8GB. All four steps of `run.sh` were run on the H200 with a hard cap of 7 GiB (`torch.cuda.set_per_process_memory_fraction`, so any allocation beyond it raises an out-of-memory error; 7 GiB leaves room for the CUDA context on an 8 GB card). Peak allocated memory: candidate generation 3.34 GiB, judge labeling 4.07 GiB (reserved 6.97 GiB at batch size 4), dense fine-tuning 4.99 GiB, reranker fine-tuning 5.85 GiB. A reranker group of 16 candidates (1,800 characters) ran out of memory under the cap, which is why the submitted recipe uses groups of 8 and 1,400 characters; the dense recipe uses mini-batches of 8 texts with gradient checkpointing. Times on the shared H200 (a 3070 will be slower; not measured): candidate generation about 6 min, judge labeling about 2 h 20 min (skipped by default, labels are downloaded), dense fine-tuning 16 min, reranker fine-tuning about 45 min (with less GPU contention the whole `run.sh`, labels step skipped, took 40 min). Running the submitted `run.sh` end to end from the unzipped submission under the 7 GiB cap and then evaluating the models it produced gave graded R@10 0.979 / MRR@10 0.895 on public, matching the 0.978 / 0.898 of the models used above. The exploration in the tables below was run before this constraint (no memory cap, larger settings); the submitted models are the capped ones. ## Strategies tried and why the submitted one is best

Exact-ID R@10 / MRR@10 unless marked graded. Single runs unless a spread is given; measured seed spread was about 0.004 (R@10) and 0.008 (MRR@10) for the settings that were repeated with 3 seeds.

#	strategy	R@10	MRR@10
1	BM25	0.765	0.595
2	zero-shot dense: bge-base / Qwen3-Emb-0.6B / bge-m3 / BLaIR-base / BLaIR-large	0.556 / 0.621 / 0.473 / 0.487 / 0.229	0.335 / 0.381 / 0.279 / 0.273 / 0.127
3	zero-shot dense + BM25 (RRF)	bge-base 0.724, Qwen3 0.762, bge-m3 0.694	0.498, 0.537, 0.462
4	#3 (Qwen3) + zero-shot reranker, top 30: MiniLM-L6 / bge-reranker-v2-m3 / Qwen2.5-1.5B (my judge-style prompt, top 20) / Qwen3-Reranker-0.6B	0.574 / 0.653 / 0.701 / 0.873	0.296 / 0.383 / 0.380 / 0.717
5	fine-tuned dense, in-batch only	0.859 (dense)	0.666
6	+ 1 hard negative; 3; 7 (lr 3e-5, 4 ep)	0.878; 0.885; 0.932 (dense)	0.694; 0.713; 0.797
7	#6 (1 hard neg) + BM25 + zero-shot Qwen3-Reranker	0.920	0.745

#	strategy	R@10	MRR@10
8	#7 with fine-tuned reranker (gold positive; or p = 0.5 alternatives; 16-group lr 2e-5)	0.938; 0.938; 0.940	0.835; 0.843; 0.857
9	**trained under the 7 GiB cap**: dense (3 negatives, 2 epochs, length 384) alone	0.919 (dense, length 384)	0.782
10	#9 + BM25 (RRF) shortlist, capped reranker (top 30)	0.948	0.856
11	#10 + query-likelihood rescoring of the reranker's top 30 / top 15 (lecture method, zero-shot)	0.959 / 0.956	0.877 / 0.876
12	#10 with pairwise reranking of the top 5 by Qwen2.5-1.5B instead (alone / combined, weight 0.25)	0.948 / 0.948	0.602 / 0.854
13	#10 with the query rewritten or HyDE-expanded before retrieval (best fused first stage)	0.953	0.846
14	dense retriever distilled from the reranker's scores instead of mined negatives (dense alone)	0.898	0.750

Judge-graded R@10 / MRR@10 of full pipelines: zero-shot hybrid 0.877 / 0.646; + zero-shot Qwen3-Reranker 0.938 / 0.803; fine-tuned dense (1 negative) + zero-shot reranker 0.960 / 0.824; same + fine-tuned reranker 0.975 / 0.883; ftF dense (3 negatives) + BM25 + R2 reranker (both trained on all data) 0.977 / 0.887; the best uncapped pipeline (7-negative 4-epoch dense, 16-candidate reranker groups) 0.986 / 0.908; **capped pipeline, trained under the 7 GiB cap (3-negative 2-epoch dense + BM25 + 8-candidate-group reranker, rerank top 30) 0.978 / 0.898; submitted pipeline with the query-likelihood stage added 0.982 / 0.917** (spec thresholds on public: Simple 0.860 / 0.650, Strong 0.885 / 0.680, Boss 0.950 / 0.800). Private metrics cannot be measured here.

Why the submitted pipeline: (a) the zero-shot dense models were all below BM25 at R@10, so lexical matching and fine-tuning both matter; fine-tuning the dense retriever on the 4,830 dialogues raised dense R@10 from 0.621 to 0.859 with in-batch negatives, 0.919 with the capped 3-negative recipe and 0.932 with an uncapped 7-negative, 4-epoch run. (b) Of the zero-shot rerankers only Qwen3-Reranker-0.6B improved the ranking (the other three lowered both exact metrics and, for MiniLM and bge-reranker, the graded ones: 0.740 / 0.410 and 0.791 / 0.494); fine-tuning it with judge-labelled negatives took exact MRR@10 on the same shortlist from 0.745 to about 0.84-0.86 (uncapped runs). (c) BM25 fusion helped in an earlier uncapped configuration (graded 0.977 / 0.887 with, 0.969 / 0.873 without, for the ftF + R2 pair; not re-measured for the capped models), but not in every dense-only comparison (for the strongest dense models, R@10 / MRR@10 was equal or lower with fusion while R@100 was higher). BLaIR (pretrained on Amazon Reviews 2023, likely overlapping this benchmark's products) was only run zero-shot as a disclosed ablation and is not used. (d) Of the lecture methods we added afterwards, only query-likelihood rescoring helped (graded 0.976 / 0.898 to 0.983 / 0.917 on the same shortlist, no training); pairwise reranking, query rewriting / HyDE and reranker distillation did not help in our settings (see "Lecture methods" below).

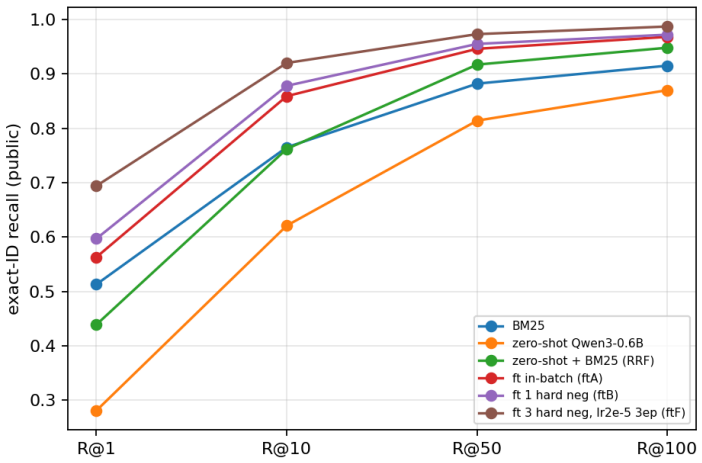


Figure 1: Exact-ID recall at k of the first-stage variants.

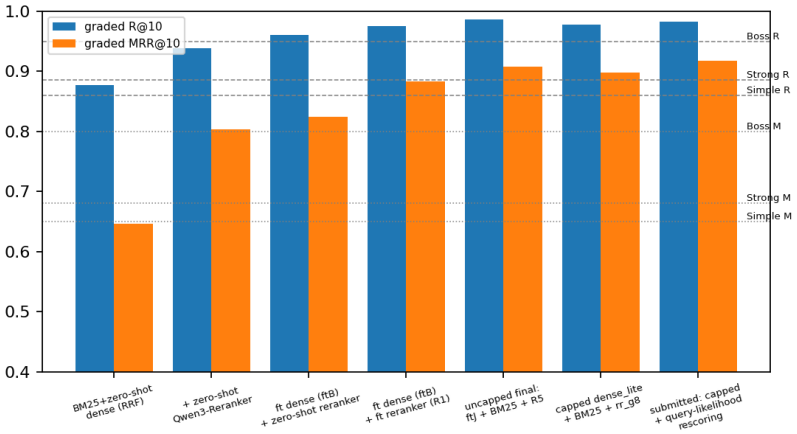


Figure 2: Graded R@10 / MRR@10 per pipeline stage, dashed lines are the spec thresholds.

Ablations and analyses

- **Rerank depth** (zero-shot shortlist, Qwen3-Reranker zero-shot, exact R@10 / MRR@10): depth 10 0.762 / 0.662, 20 0.837 / 0.700, 30 0.873 / 0.717, 50 0.886 / 0.723. First-stage exact recall@30 / @100 of the zero-shot hybrid was 0.892 / 0.948.
- **Product text per candidate at rerank time** (depth 30): 600 / 1,000 / 1,400 / 1,800 characters gave MRR@10 0.622 / 0.697 / 0.717 / 0.717.
- **Dense max length at inference** (ftD): 256 / 384 / 512 tokens gave dense R@10 0.894 / 0.907 / 0.912; end to end (graded R@10 / MRR@10) 256 tokens + depth 20 + 1,200 characters 0.970 / 0.880 in 226 s versus 512 / 30 / 1,800 0.977 / 0.891 in 412 s (ftD + R2 models).
- **Seen vs unseen gold** (submitted pipeline, exact R@10 / MRR@10): seen-in-train-pool targets (500) 0.956 / 0.879, never-seen targets (500) 0.958 / 0.876. Earlier systems also showed seen and unseen within about 2 points at R@10.
- **Per category** (200 queries each, exact R@10 / MRR@10): beauty 0.970 / 0.889, home & kitchen 0.980 / 0.931, pet supplies 0.980 / 0.867, sports & outdoors 0.960 / 0.907, toys & games 0.895 / 0.795.

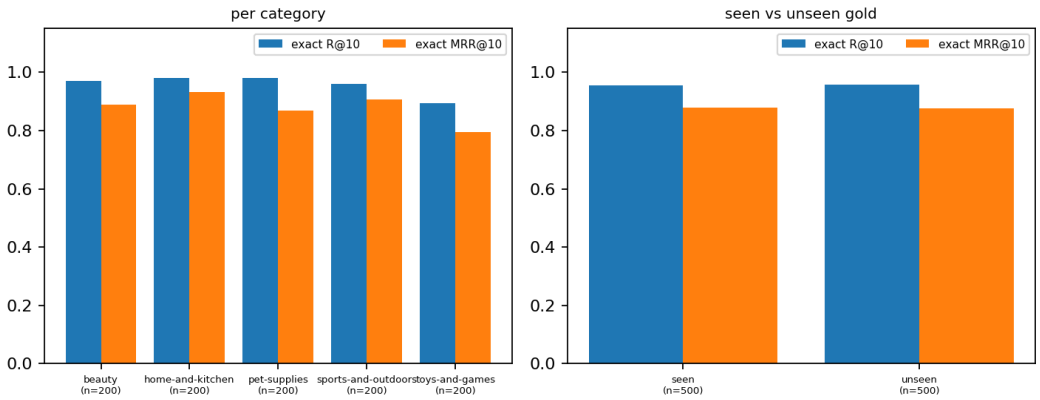


Figure 3: Submitted pipeline, per category and seen vs unseen gold (exact).

Lecture methods tried after the main pipeline

We went back through the RAG lecture for retrieval methods we had not used. All numbers are exact-ID on public, using the capped models' shortlist (BM25 + dense, RRF; recall@10 / @30 0.912 / 0.966) reranked over its top 30 (R@10 / MRR@10 0.948 / 0.856), unless noted.

- **Query likelihood** (UPR-style; slides on query-likelihood retrieval). Score = mean log P(conversation | product listing, instruction) under Qwen2.5-1.5B-Instruct, zero-shot. Alone on the shortlist: 0.940 / 0.802 (the first stage is 0.912 / 0.758). Added to the reranker as z(rerank) + w * z(ql): w = 0.1 / 0.25 / 0.5 / 1 / 2 / 3 / 5 gives MRR@10 0.861 / 0.865 / 0.871 / 0.877 / 0.866 / 0.854 / 0.845. Restricted to the reranker's top 5 / 10 / 15 / 20 (w = 1): MRR@10 0.867 / 0.873 / 0.876 / 0.876. Judge-graded on the same shortlist: 0.976 / 0.898 without, 0.983 / 0.917 with. Adopted for the top 15 (a third of the cost for nearly the same gain); the depth and weight were picked on public.
- **Pairwise reranking** (instruction-based pairwise reranking). Qwen2.5-1.5B-Instruct compares the reranker's top 5 two at a time in both orders; mean win probability per product. Alone 0.602 MRR@10, combined with the reranker (weight 0.25 / 0.5 / 1) 0.854 / 0.846 / 0.809, versus 0.856 for the reranker alone. Not helpful with this model and prompt.
- **Query rewriting and HyDE** (pre-retrieval rewriting). Qwen2.5-1.5B-Instruct rewrites the conversation into one search query or writes a hypothetical product listing. Alone, dense R@10 drops from 0.919 (original conversation) to 0.791 (rewrite) / 0.765 (HyDE). Adding the rewrite's dense list to the fusion raises first-stage recall@10 / @30 from 0.912 / 0.966 to 0.939 / 0.973, but after reranking the top 30 the result is 0.953 / 0.846 versus 0.948 / 0.856, so no gain in MRR@10.
- **Distilling the reranker into the retriever** (PairDistill idea, our own listwise variant). Teacher: the fine-tuned reranker's scores on each training dialogue's top-20 candidates; student: Qwen3-Embedding-0.6B trained with KL to the teacher softmax (temperature 2) over the gold + 15 candidates, one dialogue per step,

under the same 7 GiB cap (peak 5.64 GiB). Dense alone: R@10 0.898 / MRR@10 0.750 / R@100 0.971 versus 0.919 / 0.782 / 0.986 for the mined-negative recipe. One configuration, not tuned.

Resources

On the H200 (shared), submitted settings: `retrieval.py` 453 s (dense stage including BM25 229 s, rerank of 30,000 pairs 148 s, query likelihood of 15,000 pairs 73 s), peak allocated GPU memory 3.67 GB (graded R@10 0.982, MRR@10 0.917). The query-likelihood stage loads `Qwen/Qwen2.5-1.5B-Instruct` from the Hugging Face hub at inference time (the model the TA's judge also uses; it is not part of the downloaded tarball) and is skipped if it cannot be loaded. Training steps: see "Training fits the 8GB GPU". `evaluate.py` (judge, the TA's script) peaked at 17 GB reserved with batch size 16 on the H200 and 8.4 GB reserved with batch size 8, so the judge's memory depends on its batch size and is larger than our pipeline's. Estimate for a 3070 (interpretation, not measured): about 25-45 minutes from FLOP counting for the dense and rerank stages, plus roughly 5-10 for query likelihood, which is why the time guard exists; in a simulated 500 s budget the guard chose 256 tokens and depth about 4 and still scored 0.950 / 0.855 graded (that test predates the query-likelihood stage; with it, a simulated 600 s budget gave 0.968 / 0.893).

Q2. Generation analysis

Qwen2.5-1.5B-Instruct (bf16, greedy, 220 new tokens) answered 10 random public dialogues from the top 3 retrieved specs of the earlier uncapped final pipeline (0.986 / 0.908 graded; the submitted capped pipeline was built afterwards and not re-run for Q2) (700 characters each) under three prompts: **plain** (one-line system prompt, "write the assistant's next reply recommending a product"), **grounded** (recommend one product from the list, only facts present in the spec, mention the title and 2-3 matching features, under 80 words) and **checklist** (first list the user's firm requirements, then pick one candidate quoting its spec, then a "Not confirmed" line). The gold product was the rank-1 retrieved item in 9 of 10 dialogues. Outputs are in `results/q2_final.jsonl`.

prompt	mean words	names the rank-1 title	invented URL	"Not confirmed: None …"
plain	96	8/10	1	-
grounded	56	8/10	1	-
checklist	69	7/10	0	4

Qualitative findings (10 samples, counts are of surface features, not a hallucination rate):

1. *No-lamp French nails (beauty-000138).* plain ignored the rank-1 acrylic powder and recommended another candidate (nail tips) with details from that candidate's spec, i.e. it did not follow the best-first order; grounded and checklist recommended the retrieved top item.
2. *Deodorant stick (beauty-000065, gold at rank 2).* all three recommended the rank-1 Speed Stick, so a retrieval error flows straight into the answer; grounded also fabricated a link and a variant name ("Pack of 2") not in the specs, and checklist listed requirements the user never stated ("Cleanser", "Aluminum-free").
3. *Shaped St. Patrick's puzzle (toys-000068).* checklist wrote "Standard rectangular shape" under the requirements even though the user asked for a shaped puzzle; plain and grounded were correct and grounded correctly added the spec's age range.
4. *Hidden handgun mount (sports-000180).* checklist's "Not confirmed" line said the spec does not mention discreet mounting although the product title is "Concealed Holder", so the verification step was wrong.
5. *Whitefish cat food (pet-supplies-000108) and tablecloth clips (home-000062).* all prompts were accurate; the plain prompt added unsupported flourishes (a product URL and "trusted since 2017" for the cat food).

Interpretation (not measured beyond these samples): the short grounded prompt gave the most concise answers that stayed on spec-stated features; the checklist prompt produced structure but its verification step was empty or wrong in most samples and sometimes invented requirements; the plain prompt was longest and added the most unsupported detail. No human or judge faithfulness rating was done.

Limitations

Hyperparameters were selected using public-set metrics, so public numbers are optimistic for the chosen configuration (the capped recipe was tried once with the settings described above, after the uncapped exploration); no separate dev score was computed (a product-disjoint dev split was held out for the non-final models but not scored). The private test pool is larger (100,000 products). Most comparisons are single runs; the measured seed spread is given above. The 3070 run time is estimated, not measured.